

DATA SCIENCE

Career Roadmap 2026

Basic → Intermediate

The definitive 2026 guide for Junior & Mid-Level Data Scientists

Freshers | Career Switchers | Analytics-to-DS Transitions

8

Phases

8

Core Subjects

160+

Chapters

1,240+

Subtopics

60+

Tools

12 M

Typical Path

2026

Edition

How to Use This Roadmap

- If you come from Data Analytics — start at Phase 2. Phases 0–1 are your bridge content.
- If you are a fresh switcher or beginner — follow all 9 phases sequentially.
- Priority is law: Must-Have > High > Good-to-Have. Master Must-Haves first.
- Build one project per phase and commit it to GitHub with a proper README.
- Math & Statistics handbook (16 chapters) runs parallel — complete it alongside Phase 1.
- Python handbook (14 units) runs parallel — complete it alongside Phase 2.
- Never skip Phase 5 (Model Evaluation) — it is the most underestimated phase.
- Revisit and deepen skills iteratively — Data Science is not a linear discipline.

01. What Is Data Science in 2026?

Data Science in 2026 is a mature discipline at the intersection of statistics, software engineering, and domain expertise. The rise of LLMs, AutoML, and AI-assisted coding has shifted the role — Data Scientists now spend more time on problem framing, feature engineering, and productionising models than on manual algorithm coding. Classical ML still dominates production systems, deep learning is now a baseline expectation even at junior level, and MLOps has become a first-class requirement.

What Data Scientists Do

- Frame ambiguous business problems into ML tasks
- Source, clean, and engineer features from raw data
- Select, train, tune, and evaluate models
- Communicate findings to non-technical stakeholders
- Deploy and monitor models in production (MLOps)
- Design and analyse A/B tests and causal experiments

DS vs Data Analytics in 2026

- DA answers 'what happened' — DS answers 'what will happen'
- DA uses SQL + BI — DS uses Python + ML frameworks
- DA interprets existing data — DS builds predictive models
- DA focuses on dashboards — DS focuses on model artifacts
- DA requires statistics — DS requires statistics + ML + coding

- Build ML pipelines that run reliably at scale
- Stay current — new frameworks and papers every month

- Career path: DA → DS is a common and well-worn transition
- Salary premium: DS earns 30–60% more than DA at same level

Data Science Career Ladder — India 2026

Role	Experience	Key Responsibility	India CTC (2026)
DS Intern / Associate	0–6 months	EDA, feature engineering, model training under guidance	5–8 LPA (stipend)
Junior Data Scientist	0–2 years	Classical ML, Python, model evaluation, dashboards	8–15 LPA
Data Scientist	2–5 years	End-to-end ML, deep learning basics, MLOps, NLP	15–28 LPA
Senior Data Scientist	5–8 years	ML strategy, system design, mentoring, complex models	28–50 LPA
Principal / Staff DS	8+ years	Research direction, cross-org ML platform, patents	50–90 LPA
ML Engineer	2–6 years	Production ML systems, latency optimisation, serving	20–45 LPA
ML Research Scientist	PhD / 5+ years	Novel algorithm research, papers, model architecture	40–100 LPA+

Phase 0 | Before You Write a Single Line

Mindset, Tools & Developer Environment

The non-negotiable setup every Data Scientist must have from Day 1

The Data Scientist Mindset — Core Principles

- Every model is wrong — some are useful. Understand bias-variance before optimising anything.
- Data quality > model sophistication. A better model on dirty data loses to a simple model on clean data.
- Reproducibility is not optional. Every experiment must be fully reproducible by someone else.
- Start simple. A Logistic Regression baseline often beats complex models when features are good.
- Understand the business cost of errors. False positives and false negatives have different consequences.
- Version everything — data, code, models, experiments. Unversioned work has no production value.
- Explain your models. If you cannot explain why it predicts X, you should not deploy it.
- The last mile is the hardest. A model that cannot be deployed has zero business value.

Environment Setup

Skill / Topic	Level	Tools & Frameworks	Priority
Python 3.11+	Basic	Anaconda / pyenv / uv package manager	Must-Have
VS Code + Jupyter	Basic	VS Code, Jupyter Lab, Google Colab, Cursor IDE	Must-Have
Git & GitHub	Basic	Git CLI, GitHub, GitLab — version control	Must-Have
Virtual Environments	Basic	conda env, venv, uv venv — dependency isolation	Must-Have
Docker Basics	Intermediate	Docker Desktop, Dockerfile, docker-compose	High
Linux / Bash	Basic	WSL2, Terminal — file ops, pipes, cron	High
Markdown & Docs	Basic	VS Code, Obsidian, Notion — README writing	High
MLflow / DVC Setup	Intermediate	MLflow UI, DVC init — experiment tracking setup	High
Cloud Account	Basic	GCP Free Tier, AWS Free Tier, Azure Free	High

Phase 1 | The Language of Data Science

Mathematics & Statistics

Non-negotiable foundation — 2-part subject running parallel to this roadmap

1.1 Statistics for Data Scientists — Core Topics

Skill / Topic	Level	Tools & Frameworks	Priority
Descriptive Statistics	Basic	pandas describe(), scipy.stats — central tendency, dispersion	Must-Have
Probability Foundations	Basic	numpy, scipy.stats — events, distributions, Bayes rule	Must-Have
Probability Distributions	Basic	Normal, Binomial, Poisson, t, F, Chi-Sq — scipy.stats	Must-Have
Normal Distribution	Basic	scipy.stats.norm — PDF, CDF, Z-scores	Must-Have
Central Limit Theorem	Basic	numpy simulation — sampling distributions	Must-Have
Hypothesis Testing	Basic	scipy.stats — z-test, t-test, chi-sq, ANOVA	Must-Have
Confidence Intervals	Basic	scipy.stats, statsmodels — interval estimation	Must-Have
A/B Testing	Intermediate	scipy.stats — experiment design, significance testing	Must-Have
Correlation & Regression	Basic	pandas, scipy.stats, seaborn — Pearson, Spearman, OLS	Must-Have
Bayes Theorem Basics	Basic	Manual calculation, scipy — conditional probability	Must-Have
Analysis of Variance	Intermediate	scipy.stats — one-way ANOVA, post-hoc tests	Must-Have
Time Series Statistics	Intermediate	statsmodels — ARIMA, Prophet, ACF/PACF	Must-Have

1.2 Mathematics for Machine Learning

Skill / Topic	Level	Tools & Frameworks	Priority
Vectors, Matrices & Operations	Basic	numpy — dot, matmul, transpose, broadcasting	Must-Have
Matrix Basics & Inverse	Basic	numpy.linalg — inv, det, solve — OLS normal equation	Must-Have

Skill / Topic	Level	Tools & Frameworks	Priority
Eigenvalues & Eigenvectors	Intermediate	numpy.linalg.eig — PCA intuition, variance decomposition	Must-Have
Cosine Similarity	Basic	numpy, sklearn.metrics — NLP, embeddings, RAG retrieval	Must-Have
Derivatives & Chain Rule	Basic	sympy, conceptual — loss minimisation, backprop foundation	Must-Have
Partial Derivatives & Gradient Descent	Intermediate	numpy, autograd — gradient of loss w.r.t. parameters	Must-Have
Convexity & Loss Functions	Intermediate	MSE, Cross-Entropy, Hinge — convex vs. non-convex loss	Must-Have

Phase 2 | Your Primary Toolkit

Python for Data Science

Master the scientific Python stack — from raw arrays to production-ready code

2.1 Scientific Python Stack — Advanced

Skill / Topic	Level	Tools & Frameworks	Priority
SciPy — Scientific Computing	Intermediate	scipy.stats, scipy.optimize, scipy.spatial, scipy.signal	High
Plotly — Advanced Interactive Viz	Intermediate	plotly.graph_objects, make_subplots, 3D plots, ML charts	High
FastAPI & API Integration	Intermediate	fastapi, uvicorn, httpx, asyncio, pydantic — API design	High
Advanced Python for DS	Intermediate	OOP, generators, decorators, context managers, pathlib	High

2.2 Data Storage & Retrieval

Skill / Topic	Level	Tools & Frameworks	Priority
SQL in Python	Basic	SQLAlchemy, psycopg2, sqlite3, pandas read_sql	Must-Have
NoSQL — MongoDB Basics	Intermediate	pymongo — document store, CRUD, aggregation pipeline	Good-to-Have
Cloud Data — BigQuery	Intermediate	google-cloud-bigquery, pandas-gbq — warehouse queries	High
Parquet & Arrow	Intermediate	pyarrow, pandas — columnar storage for large data	High
DuckDB	Intermediate	duckdb — in-process analytics SQL on CSV/Parquet files	High

2.3 Code Quality & Engineering Practices

Skill / Topic	Level	Tools & Frameworks	Priority
Python Packaging	Intermediate	pyproject.toml, pip install -e ., virtual environments	High
Testing ML Code	Intermediate	pytest — data pipelines, preprocessing, model output tests	High

Skill / Topic	Level	Tools & Frameworks	Priority
Logging & Error Handling	Intermediate	Python logging, structlog — production structured logging	High
Type Hints, Docstrings & Linting	Intermediate	mypy, ruff, pre-commit — type safety and code quality	High

Phase 3 | Where Real DS Work Begins

EDA & Feature Engineering

80% of a Data Scientist's time is here — better features always beat better algorithms

3.1 Exploratory Data Analysis

Skill / Topic	Level	Tools & Frameworks	Priority
Data Types & Feature Understanding	Basic	pandas — dtypes, cardinality, variable roles, data dictionary	Must-Have
Data Quality Assessment	Basic	pandas — duplicates, schema validation, quality scorecard	Must-Have
Automated EDA Profiles	Basic	ydata-profiling, sweetviz, dtale — one-click EDA reports	Must-Have
Univariate Analysis	Basic	pandas, seaborn — distributions, skewness, kurtosis	Must-Have
Bivariate Analysis	Basic	matplotlib, seaborn — scatter, heatmaps, pair plots	Must-Have
Multivariate Analysis	Intermediate	PCA plots, feature importance, parallel coordinates	Must-Have
Missing Data Analysis	Intermediate	missingno — MCAR/MAR/MNAR patterns, visualisation	Must-Have
Outlier Detection	Intermediate	IQR, z-score, IsolationForest, LOF — detection methods	Must-Have
Class Imbalance Analysis	Intermediate	imbalanced-learn, SMOTE — value counts, distribution check	Must-Have
Target Leakage Detection	Intermediate	correlation analysis, time-aware splits — leakage audit	Must-Have

3.2 Feature Engineering

Skill / Topic	Level	Tools & Frameworks	Priority
Numerical Transformations	Basic	log, sqrt, Box-Cox, quantile transform — sklearn	Must-Have
Encoding Categorical Variables	Basic	LabelEncoder, OneHotEncoder, OrdinalEncoder — sklearn	Must-Have
Target Encoding	Intermediate	category_encoders — mean target per category, smoothing	Must-Have

Skill / Topic	Level	Tools & Frameworks	Priority
Feature Scaling & Normalisation	Basic	StandardScaler, MinMaxScaler, RobustScaler — sklearn	Must-Have
Date/Time Feature Extraction	Intermediate	pandas dt accessor — hour, day_of_week, quarter, lag	Must-Have
Interaction & Polynomial Features	Intermediate	sklearn PolynomialFeatures, manual feature crosses	High
Binning & Discretisation	Intermediate	pandas cut/qcut, KBinsDiscretizer, optimal binning	High
Text Feature Extraction	Intermediate	TF-IDF, CountVectorizer, sklearn Pipeline — tabular NLP	Must-Have
Aggregation Features	Intermediate	groupby agg, lag features, rolling stats for time series	Must-Have
High Cardinality Features	Intermediate	frequency encoding, hashing, entity embeddings — rare cats	Must-Have
Dimensionality Reduction	Intermediate	PCA, t-SNE, UMAP — sklearn, umap-learn	High
Domain Knowledge Feature Creation	Intermediate	RFM features, business ratios, composite scores	Must-Have
Feature Selection	Intermediate	filter methods, RFE, Lasso, permutation importance	Must-Have

3.3 Data Cleaning & Preprocessing

Skill / Topic	Level	Tools & Frameworks	Priority
Handling Missing Values	Basic	SimpleImputer, KNNImputer, IterativeImputer — sklearn	Must-Have
Handling Outliers	Intermediate	Capping, Winsorising, IsolationForest — treatment strategies	Must-Have
Handling Inconsistent & Erroneous Data	Basic	pandas str methods, regex, cross-column validation	Must-Have
Handling Duplicates	Basic	pandas duplicated, drop_duplicates, fuzzy matching	Must-Have
Data Type Conversion & Formatting	Basic	pandas astype, to_datetime, category dtype — dtype fixes	Must-Have
sklearn Pipeline	Intermediate	Pipeline, ColumnTransformer — leak-free preprocessing	Must-Have

Phase 4 | The Core of Data Science

Classical Machine Learning

From Linear Regression to Gradient Boosting — every algorithm a DS must know cold

4.1 Supervised Learning — Regression

Skill / Topic	Level	Tools & Frameworks	Priority
Linear Regression (OLS)	Basic	sklearn LinearRegression, statsmodels — baseline regressor	Must-Have
Ridge & Lasso Regression	Intermediate	sklearn Ridge, Lasso, ElasticNet — L1/L2 regularisation	Must-Have
Polynomial Regression	Intermediate	sklearn PolynomialFeatures + LinearRegression	High
Decision Tree Regressor	Basic	sklearn DecisionTreeRegressor — splitting, pruning	Must-Have
Random Forest Regressor	Intermediate	sklearn RandomForestRegressor — bagging ensemble	Must-Have
Gradient Boosting — XGBoost	Intermediate	xgboost XGBRegressor — industry standard boosting	Must-Have
Gradient Boosting — LightGBM	Intermediate	lightgbm LGBMRegressor — faster than XGBoost	Must-Have
CatBoost Regressor	Intermediate	catboost CatBoostRegressor — native categorical handling	High
SVR — Support Vector Regression	Intermediate	sklearn SVR — kernel trick for regression	Good-to-Have

4.2 Supervised Learning — Classification

Skill / Topic	Level	Tools & Frameworks	Priority
Logistic Regression	Basic	sklearn LogisticRegression — always the baseline classifier	Must-Have
Naive Bayes	Basic	sklearn GaussianNB, MultinomialNB — probabilistic classifier	Must-Have
K-Nearest Neighbours (KNN)	Basic	sklearn KNeighborsClassifier — instance-based learning	Must-Have
Decision Tree Classifier	Basic	sklearn DecisionTreeClassifier — Gini / entropy splits	Must-Have

Skill / Topic	Level	Tools & Frameworks	Priority
Random Forest Classifier	Intermediate	sklearn RandomForestClassifier — class_weight, OOB score	Must-Have
XGBoost Classifier	Intermediate	xgboost XGBClassifier — Kaggle standard, SHAP support	Must-Have
CatBoost Classifier	Intermediate	catboost CatBoostClassifier — native categorical support	High
LightGBM Classifier	Intermediate	lightgbm LGBMClassifier — fast, leaf-wise growth	Must-Have
SVM — Support Vector Machine	Intermediate	sklearn SVC, LinearSVC — max margin classifier	High
Stacking & Voting Ensembles	Intermediate	sklearn StackingClassifier, VotingClassifier	High

4.3 Unsupervised Learning

Skill / Topic	Level	Tools & Frameworks	Priority
K-Means Clustering	Basic	sklearn KMeans — elbow method, silhouette score	Must-Have
Hierarchical Clustering	Intermediate	sklearn AgglomerativeClustering, scipy dendrogram	High
DBSCAN	Intermediate	sklearn DBSCAN — density-based, handles noise, no K needed	High
t-SNE & UMAP	Intermediate	sklearn TSNE, umap-learn — 2D/3D cluster visualisation	Must-Have
Anomaly Detection	Intermediate	IsolationForest, LOF, One-Class SVM — outlier modeling	High
Association Rules	Intermediate	mlxtend Apriori — market basket analysis	Good-to-Have

Phase 5 | Know If Your Model Actually Works

Model Evaluation, Selection & Tuning

The most underestimated phase — a model you cannot evaluate is a model you cannot trust

5.1 Evaluation Metrics

Skill / Topic	Level	Tools & Frameworks	Priority
Regression Metrics	Basic	MAE, MSE, RMSE, MAPE, R^2 , Adjusted R^2 — sklearn metrics	Must-Have
Classification — Threshold Metrics	Basic	Accuracy, Precision, Recall, F1, Confusion Matrix	Must-Have
AUC-ROC Curve	Basic	sklearn roc_auc_score, roc_curve — threshold-independent	Must-Have
Precision-Recall Curve	Intermediate	sklearn average_precision_score — better for imbalance	Must-Have
Log Loss (Cross-Entropy)	Intermediate	sklearn log_loss — probabilistic classifier evaluation	Must-Have
Multi-class Metrics	Intermediate	macro/micro/weighted F1, cohen_kappa_score	Must-Have
Business-Aligned Metrics	Intermediate	Custom loss, cost matrix, lift/gain charts, expected value	High
Clustering Metrics	Intermediate	Silhouette, Davies-Bouldin, Adjusted Rand Index	High

5.2 Validation Strategies

Skill / Topic	Level	Tools & Frameworks	Priority
Train / Validation / Test Split	Basic	sklearn train_test_split — 60/20/20 or 70/15/15	Must-Have
k-Fold Cross-Validation	Basic	sklearn KFold, cross_val_score, cross_validate	Must-Have
Stratified k-Fold	Intermediate	StratifiedKFold — preserving class proportions per fold	Must-Have
Time Series CV (Walk-Forward)	Intermediate	sklearn TimeSeriesSplit — no future leakage in TS data	Must-Have
Nested Cross-Validation	Intermediate	Unbiased evaluation + tuning — outer/inner loop setup	High
Hold-out Set Best Practices	Intermediate	One-touch rule — never use test set until final eval	Must-Have

Skill / Topic	Level	Tools & Frameworks	Priority
Bias-Variance Tradeoff & Learning Curves	Intermediate	sklearn learning_curve, validation_curve — diagnosis	Must-Have

5.3 Hyperparameter Tuning & Model Debugging

Skill / Topic	Level	Tools & Frameworks	Priority
GridSearchCV	Basic	sklearn GridSearchCV — exhaustive search, small param grids	Must-Have
RandomizedSearchCV	Intermediate	sklearn RandomizedSearchCV — faster, large param spaces	Must-Have
Bayesian Optimisation (Optuna)	Intermediate	optuna — state-of-the-art HPO, pruning, visualisation	Must-Have
Hyperopt	Intermediate	hyperopt — TPE algorithm, SparkTrials for distributed HPO	Good-to-Have
Early Stopping	Intermediate	XGBoost/LightGBM/CatBoost early_stopping_rounds	Must-Have
Model Selection & Comparison	Intermediate	Leaderboard tables, statistical comparison, Occam's Razor	Must-Have
SHAP & Feature Importance	Intermediate	shap — TreeExplainer, summary plot, waterfall, force plot	Must-Have
Residual Analysis	Intermediate	Residual plots, Q-Q plots, Cook's distance, calibration	Must-Have
LIME	Intermediate	lime — LimeTabularExplainer, local model-agnostic explanation	Good-to-Have

Phase 6 | Neural Networks & Deep Learning

Deep Learning Fundamentals

A 2026 baseline expectation — you need this even for mid-level DS roles

6.1 Neural Network Foundations

Skill / Topic	Level	Tools & Frameworks	Priority
Perceptron & MLP Architecture	Basic	sklearn MLPClassifier, manual numpy implementation	Must-Have
Activation Functions	Basic	ReLU, sigmoid, tanh, GELU, softmax — torch.nn	Must-Have
Loss Functions	Basic	MSE, BCE, CCE, Huber — nn.MSELoss, nn.CrossEntropyLoss	Must-Have
Backpropagation	Intermediate	Chain rule intuition — .backward(), autograd computation	Must-Have
Optimisers	Intermediate	SGD, Momentum, Adam, AdamW — torch.optim	Must-Have
Regularisation in DL	Intermediate	Dropout, L2 weight decay, BatchNorm, LayerNorm	Must-Have
Learning Rate Schedules	Intermediate	StepLR, CosineAnnealing, OneCycleLR, ReduceLROnPlateau	High

6.2 PyTorch — Primary Framework (2026)

Skill / Topic	Level	Tools & Frameworks	Priority
PyTorch Tensors & Autograd	Basic	torch.Tensor, requires_grad, .backward(), .detach()	Must-Have
nn.Module — Building Models	Basic	torch.nn — Linear, Conv2d, LSTM, Embedding, Sequential	Must-Have
DataLoader & Dataset	Intermediate	torch.utils.data — custom Dataset, batching, WeightedSampler	Must-Have
Training Loop (Manual)	Intermediate	zero_grad, forward, loss, backward, optimizer.step()	Must-Have
PyTorch Lightning	Intermediate	pytorch-lightning — LightningModule, Trainer, callbacks	High
Model Saving & Loading	Intermediate	torch.save, torch.load, state_dict, ONNX export	Must-Have
GPU Training	Intermediate	.to(device), CUDA, mixed precision, Google Colab GPU	High

6.3 Computer Vision Basics

Skill / Topic	Level	Tools & Frameworks	Priority
Convolutional Neural Networks	Intermediate	Conv2d, MaxPool2d, torchvision — CNN architecture	Must-Have
Transfer Learning	Intermediate	torchvision ResNet, EfficientNet — feature extract / fine-tune	Must-Have
Image Augmentation	Intermediate	torchvision.transforms, albumentations — train-only augment	High
Object Detection — YOLO Basics	Intermediate	ultralytics YOLOv8 — inference, custom fine-tuning	Good-to-Have

6.4 Natural Language Processing Basics

Skill / Topic	Level	Tools & Frameworks	Priority
Text Preprocessing for Deep Learning	Basic	HuggingFace tokenizers — BPE, padding, attention mask	Must-Have
Word Embeddings	Intermediate	Word2Vec, GloVe, fastText, gensim, nn.Embedding	Must-Have
Transformer Architecture	Intermediate	Attention, self-attention, multi-head, positional encoding	Must-Have
HuggingFace Transformers	Intermediate	AutoModel, AutoTokenizer, Pipeline API, datasets library	Must-Have
Fine-Tuning Pre-trained Models	Intermediate	TrainingArguments, Trainer API, BERT fine-tuning	Must-Have
LLM APIs (GPT, Gemini, Claude)	Intermediate	openai, anthropic, google-generativeai — API consumption	High
Retrieval-Augmented Generation	Intermediate	LangChain, FAISS, ChromaDB — document QA pipeline	High

Phase 7 | Models in the Real World

MLOps & Model Deployment

A model that cannot be deployed has zero business value — this phase is non-optional

7.1 From Experiment to Deployment

Skill / Topic	Level	Tools & Frameworks	Priority
MLflow — Experiment Tracking	Intermediate	mlflow.log_param, log_metric, log_artifact, autolog	Must-Have
MLflow Model Registry	Intermediate	register_model, staging/production transitions, pyfunc load	Must-Have
DVC — Data Version Control	Intermediate	dvc add, push, pull, pipelines, dvc repro	High
Weights & Biases (wandb)	Intermediate	wandb.init, log, Artifact, Sweeps — deep learning tracking	High
FastAPI — Model Serving	Intermediate	FastAPI, uvicorn, Pydantic — REST prediction endpoint	Must-Have
Streamlit — Model Demo Apps	Intermediate	st.cache_resource, prediction UI, Streamlit Cloud deploy	Must-Have
Gradio	Intermediate	gr.Interface, gr.Blocks — HuggingFace Spaces demos	High
BentoML	Intermediate	bentoml.save_model, Service, bentoml build — ML serving	Good-to-Have
ONNX — Model Portability	Intermediate	torch.onnx.export, sklearn-onnx, onnxruntime inference	Good-to-Have
Docker — Containerising Models	Intermediate	Dockerfile, docker build/run, docker-compose, slim images	Must-Have
Cloud Run / App Engine	Intermediate	gcloud run deploy — serverless container ML API on GCP	High
AWS SageMaker Basics	Intermediate	Notebooks, Training Jobs, Endpoints, Batch Transform	High
Google Vertex AI Basics	Intermediate	Workbench, Training, Model Registry, Endpoints, Pipelines	High
CI/CD for ML (GitHub Actions)	Intermediate	workflows, pytest in CI, Docker build, Cloud Run deploy	High

7.2 Production Operations

Skill / Topic	Level	Tools & Frameworks	Priority
ML Pipelines & Workflow Orchestration	Intermediate	prefect, apache-airflow — scheduled retraining pipelines	Must-Have
Data Drift Monitoring	Intermediate	evidently — DataDriftPreset, PSI, KS test, alerts	Must-Have
Model Performance Monitoring	Intermediate	evidently ClassificationPreset — AUC/RMSE over time	Must-Have
Logging Predictions	Intermediate	structlog, PostgreSQL — input, output, model version, PII	High
Shadow Deployment	Intermediate	Parallel model serving — compare shadow vs. production	High
A/B Testing Models in Production	Intermediate	Champion-challenger pattern — traffic split, significance	High

Phase 8 | The 2026 Frontier

Generative AI & LLM Integration

Every DS must understand how to work with and build on top of LLMs

8.1 Building with LLMs

Skill / Topic	Level	Tools & Frameworks	Priority
Advanced Prompt Engineering	Intermediate	Chain-of-Thought, ReAct, structured output, prompt chaining	Must-Have
Vector Databases	Intermediate	FAISS, ChromaDB, Pinecone, Qdrant — ANN search, metadata filter	Must-Have
Advanced RAG Patterns	Intermediate	Hybrid search, re-ranking, contextual compression, RAGAS eval	Must-Have
LLM Evaluation	Intermediate	RAGAS, G-Eval, LLM-as-judge, benchmark datasets	High
Fine-Tuning Small LLMs	Intermediate	LoRA, QLoRA, HuggingFace PEFT, SFTTrainer — trl library	High
LLM Observability	Intermediate	LangSmith, Helicone — tracing, cost tracking, quality alerts	Good-to-Have
LLM Security & Safety	Intermediate	Prompt injection, guardrails-ai, PII redaction, output filtering	High
LLM APIs — Advanced Usage	Intermediate	Streaming, function calling, async, batch API, cost routing	Must-Have

8.2 AI-Augmented Data Science Workflow

Skill / Topic	Level	Tools & Frameworks	Priority
AI-Assisted Coding	Basic	GitHub Copilot, Cursor IDE, Claude Code — daily DS workflow	Must-Have
AI for EDA	Intermediate	PandasAI, Julius AI, Marimo — NL-driven data analysis	High
AutoML Tools	Intermediate	AutoGluon, H2O AutoML, FLAML, Google AutoML — fast baseline	High
Synthetic Data Generation	Intermediate	SDV, CTGAN — privacy-safe tabular and text data generation	Good-to-Have
AI Code Review	Intermediate	LLM-assisted review, refactoring, docstring generation	High

Choose Your Direction

Specialisation Tracks

Pick ONE after completing Phases 0–8 — depth beats breadth for career progression

Track 1: ML Engineering

- Production ML systems at scale
- C++/Rust for model inference speed
- Kubernetes + Kubeflow
- Model quantisation & pruning
- Feature stores (Feast, Hopsworks)
- Stream ML (Kafka + Flink)
- Target roles: ML Engineer, Platform DS

Track 2: NLP / LLM Specialist

- Transformer architecture depth
- Fine-tuning LLMs (LoRA, QLoRA)
- RAG system design & optimisation
- LLM evaluation & red-teaming
- Agents (LangGraph, AutoGen)
- LLM deployment & cost optimisation
- Target roles: NLP Engineer, LLM Engineer

Track 3: Computer Vision

- CNN architectures depth (ResNet, ViT)
- Object detection (YOLOv9+, DETR)
- Segmentation (SAM, Mask R-CNN)
- Image generation (Diffusion models)
- Video understanding
- Edge deployment (TensorRT, ONNX)
- Target roles: CV Engineer, Robotics DS

Track 4: Time Series & Analytics DS

- Advanced ARIMA, SARIMA, SARIMAX
- Neural forecasting (N-BEATS, TFT)
- Causal inference (DoWhy, EconML)
- Experiment design (Bayesian)
- Geospatial analytics
- Finance/supply chain domain depth
- Target roles: Quant DS, Supply Chain DS

Track 5: Healthcare / Life Sciences

- Clinical trial design & analysis
- Survival analysis (lifelines, KM)
- Medical imaging (MONAI, NiftI)
- EHR data modelling (HL7, FHIR)
- Regulatory & compliance awareness
- Genomics basics (bioinformatics)
- Target roles: Clinical DS, BioML Engineer

Track 6: Recommender Systems

- Collaborative filtering (ALS, SVD)
- Content-based filtering
- Deep learning for RecSys (NCF, DIN)
- Session-based recommendation
- Two-tower models (YouTube, TikTok arch)
- Feature stores & embedding serving
- Target roles: RecSys Engineer, E-Commerce DS

02. Recommended 10–12 Month Learning Timeline

Assumes 4–6 hours of focused study daily. Career switchers with programming backgrounds can compress Phases 0–2. Absolute beginners should expect 6–12 months. Adjust to your situation — pace matters less than consistency.

Period	Theme	Focus Areas
Month 1	Foundation	Math & Stats handbook + Python handbook + EDA basics + Git & Environment setup
Month 2	Core ML	Classical ML all algorithms + Feature Engineering + Model Evaluation + 1st Kaggle competition
Month 3	Advanced ML	Ensembles + Optuna tuning + SHAP/XAI + Full sklearn Pipeline project on real dataset
Month 5	Deep Learning	PyTorch foundations + CNNs + NLP + HuggingFace fine-tuning + DL capstone project
Month 6	MLOps & GenAI	MLflow + DVC + FastAPI + Docker + LLMs + RAG application build
Month 7	Job Ready	Portfolio polish + Mock interviews + Kaggle Top-30% + Apply to 60+ roles + Network

Projects to Build Per Phase

- Phase 2–3: End-to-End EDA + Feature Engineering — House Prices or Titanic (Kaggle classic, full pipeline)
- Phase 4: Classification Benchmark — churn prediction, fraud detection, or medical diagnosis with SHAP report
- Phase 5: Full ML Pipeline — sklearn Pipeline + Optuna HPO + SHAP explainability + model comparison report
- Phase 6: Deep Learning Project — sentiment classifier with BERT fine-tuning deployed on HuggingFace Spaces
- Phase 7: MLOps End-to-End — FastAPI serving + MLflow tracking + Docker containerisation + GitHub Actions CI
- Phase 8: RAG Application — document Q&A chatbot with vector DB, LLM API, and RAGAS evaluation
- Final Portfolio: 4–5 GitHub projects + 1 Kaggle Top-30% notebook + 1 technical blog post (Medium/Substack)

03. Job-Ready Checklist (2026)

Before sending your first application, verify every item below. This reflects what hiring managers at top-tier companies actually look for in Junior and Mid-Level Data Scientist candidates in 2026.

✓	Milestone	Evidence / Artifact
☐	Implement 10+ algorithms from scratch in numpy (no sklearn)	GitHub repository with README
☐	Complete 3+ end-to-end ML projects with SHAP explainability	GitHub + Jupyter notebooks
☐	Achieve Top 30% on at least one Kaggle competition	Kaggle profile URL
☐	Build and deploy one ML model as a FastAPI + Docker API	Live URL or GitHub repo
☐	Track experiments with MLflow on a real project	Screenshot in portfolio
☐	Fine-tune a HuggingFace model on a custom dataset	GitHub / HuggingFace model card
☐	Build a RAG application with a vector database	GitHub repo with demo
☐	Explain bias-variance, regularisation, and boosting without notes	Practice with a peer
☐	Complete ML System Design mock interview (recommender / fraud)	Pramp or Exponent platform
☐	Write 2 technical blog posts on ML topics	Published article URLs
☐	LinkedIn profile: DS-focused headline, featured projects, 5 skills	linkedin.com/in/yourprofile
☐	Resume: 1 page, quantified bullet points, tailored to each JD	PDF resume
☐	Apply to 60+ roles and track all in a structured spreadsheet	Job tracker sheet

*"The goal of a Data Scientist is not to build the most sophisticated model.
It is to solve the right problem, reliably, in production."*

9 Phases · 8 Core Subjects · 160+ Chapters · 1,240+ Subtopics · 6-12 Months
Start with Phase 0. Commit to GitHub daily. Ship something every month.